



## Chapter 3

# DESIGN AND DEPLOYMENT OF EMBEDDED SYSTEMS

1



- System design
  - Identify requirements
  - Specification
  - Hardware and software partitioning
  - System design
- Installation and testing embedded system

2



### Identify requirements

- Classic design : Hardware and software are separated
- Now : both hardware and software have to be calculated carefully during designing them
- Therefore this approach is called hardware – software codesign
- Available Components, which is used in design, must be listed for system
- System as simple as possible
- Strict time requirements bring products to customers
- Ability to reused of product

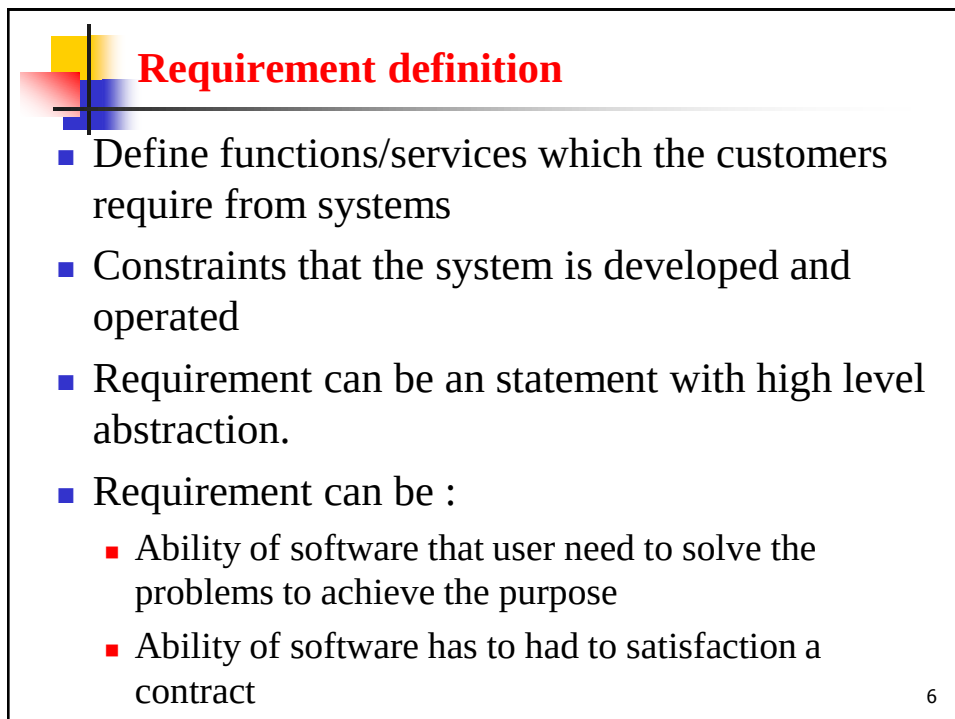
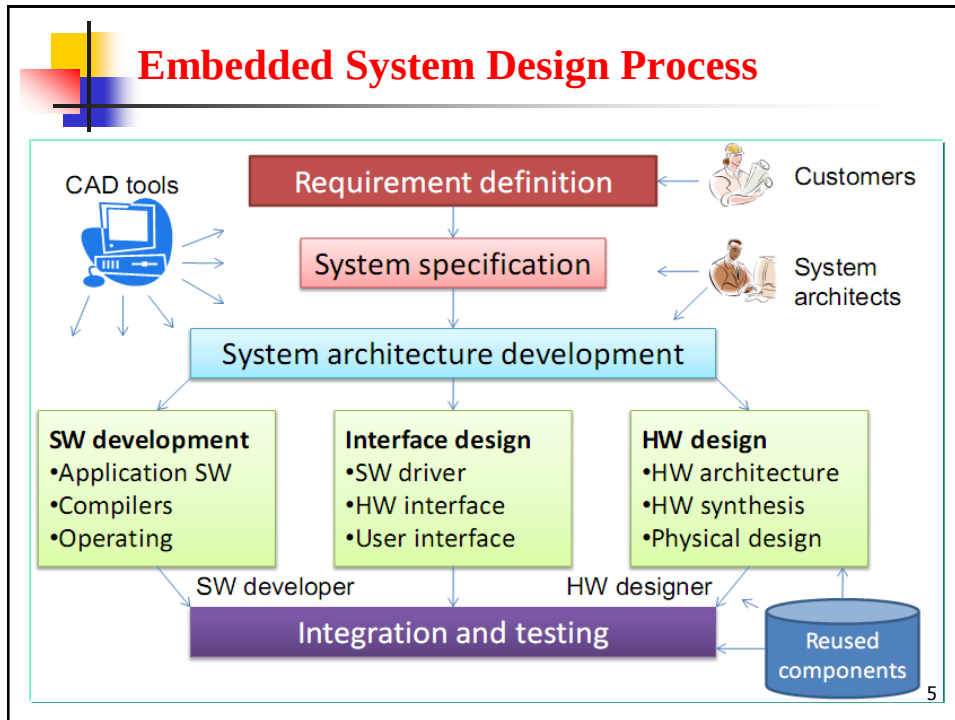
3

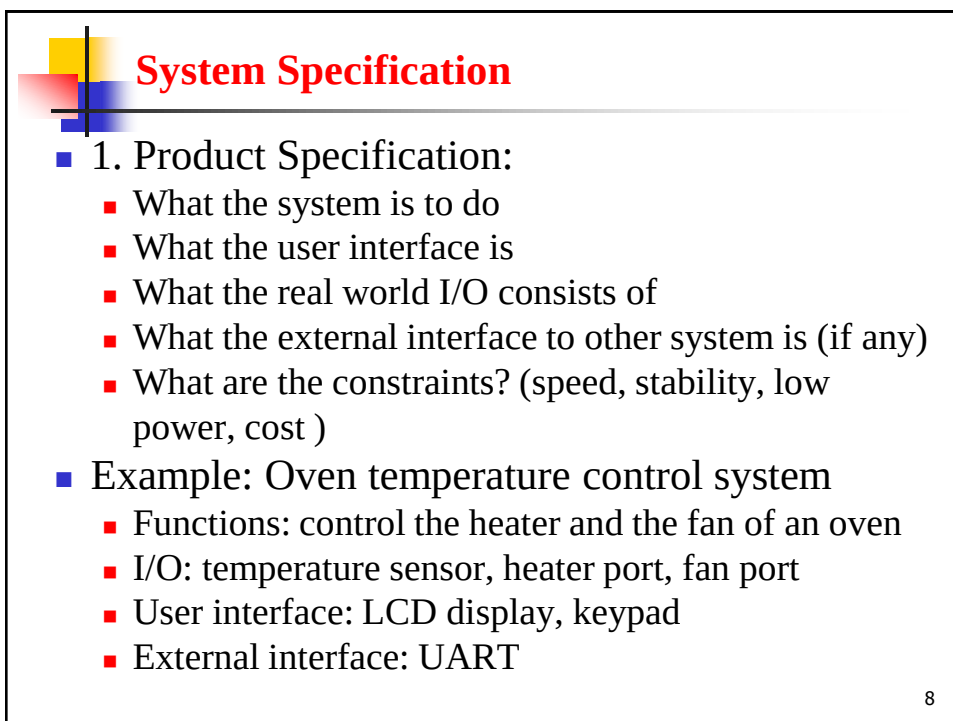
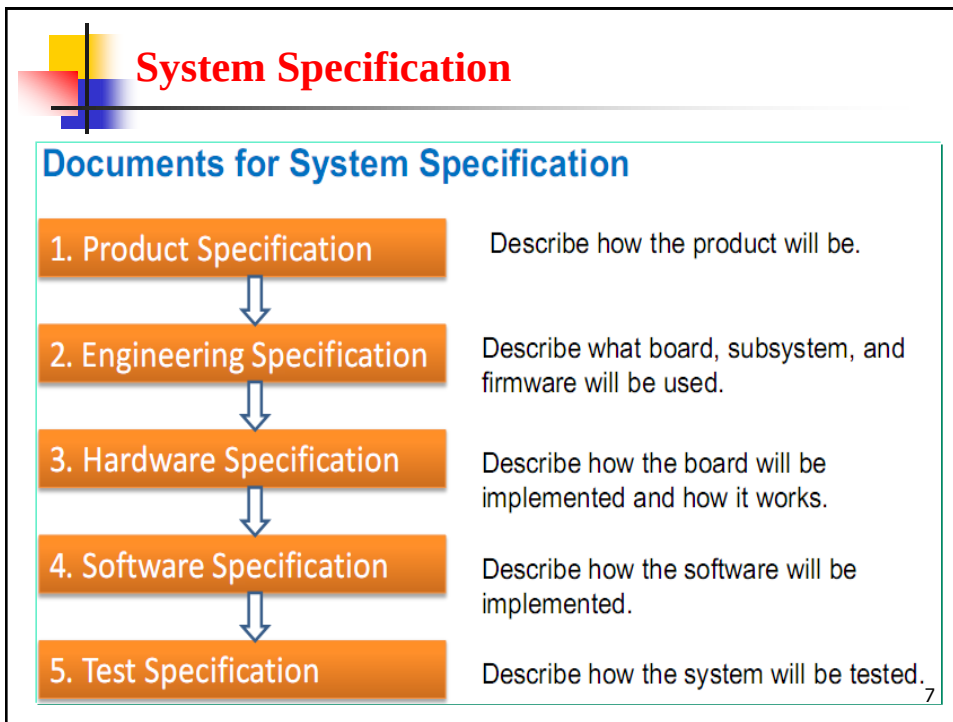


### Platform-based design (TK trên nền)

- Platform is an architecture family, which satisfying the set of constraints allow reuse the software/hardware components
- Often requires the use of a platform application programming interface (API)
- Is an approach meet-in-the-middle

4







## System Specification

- 2. Engineering Specification:
  - What kind of hardware will be used
  - What are the requirements for hardware and software
- Example: Oven temperature control system
  - 8051 microcontroller, LM35 sensor, LCD 16x2-B, ADC0809, RS232 IC
  - Requirements
    - PID control algorithm
    - real-time processing
    - Display current temperature value
    - able to set operating temperature value
    - Transfer data to computer through UART

9



## System Specification

- 3. Hardware Specification:
  - The requirements from engineering documents
  - How the hardware implements the functionality
  - The software interfaces to the hardware
- Example: Oven temperature control system
  - PIC Microcontroller 12MHz, sensor LM35, LCD 16x2-B, Keypad 16, ADC0809, FET IRF260 for heater/fan control
  - Microcontroller reads temperature value from LM35 through ADC, display this value to LCD, and then control heater and fan based on PID control algorithm

10



## System Specification

- 4. Software Specification:
  - The requirements from engineering specification
  - Interface to other software
  - How the software implements the requirements
- Example: Oven temperature control system
  - design a function:
    - `void LCD_display(String str)`
    - `int PID_control(int temp, int data[])`

11



## Test Specification

- 5. Test Specification:
  - The device, equipment, environment for testing
  - Prototype
  - Testing process
- Example: Oven temperature control system
  - Voltage meter, temperature meter
  - Prototype: bread board
  - Testing process:
    - calibrate temperature sensor
    - check LCD, keypad
    - check output port (heater, fan)
    - verify PID control algorithm

12



## System architecture development

- Partitioning the system into three parts:
  - Interface design
  - Hardware design
  - Software development
- Methodology
  - Block diagram
  - Waveform
  - Function description
  - Coding guideline

13



## System architecture development

- Hardware block diagram
  - Use a rectangle for a hardware block



- Use an arrow for a connection



- Use a symbol for a special block



Speaker



Lamp



Energy



Network

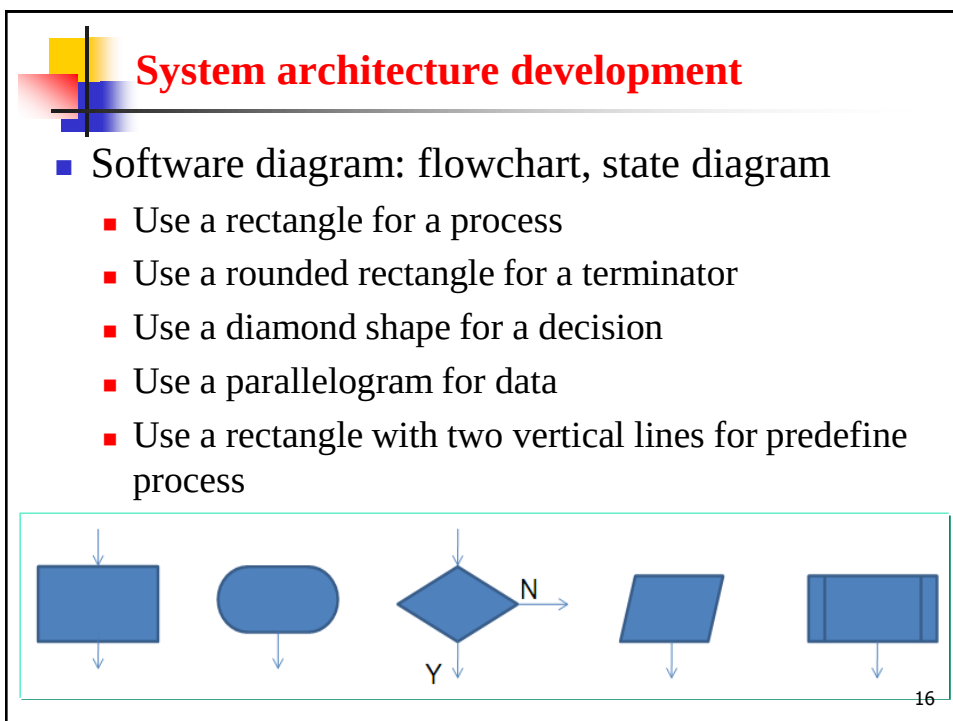
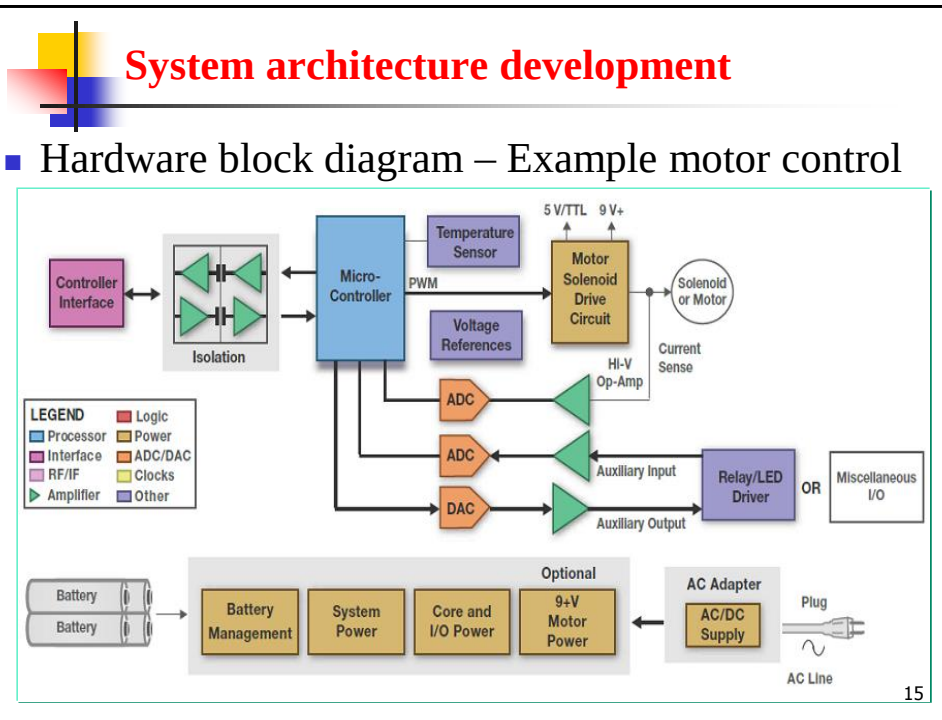


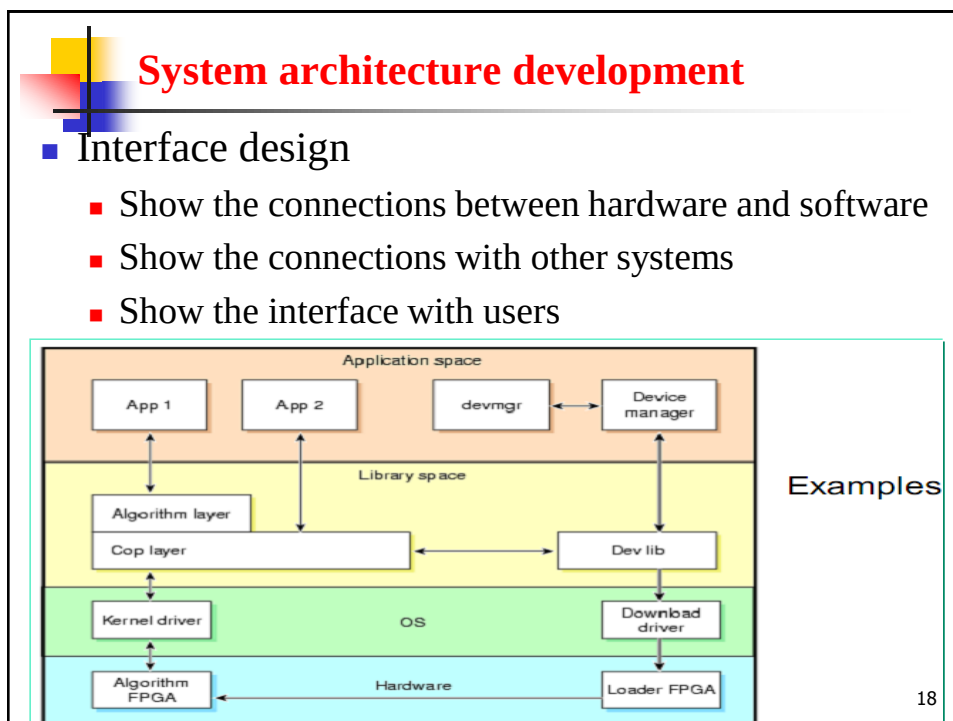
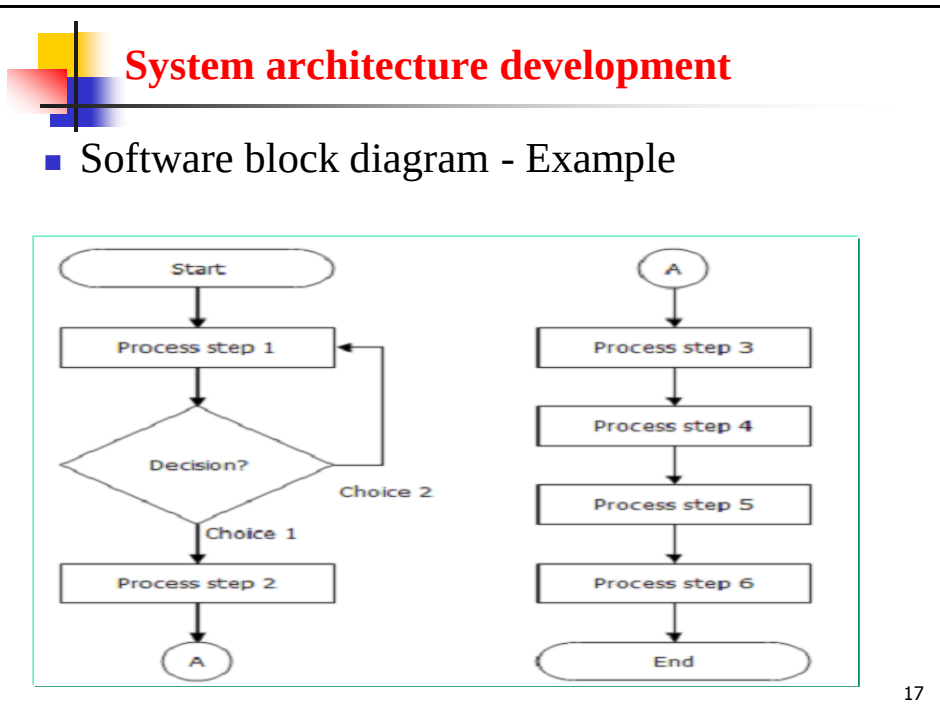
computer

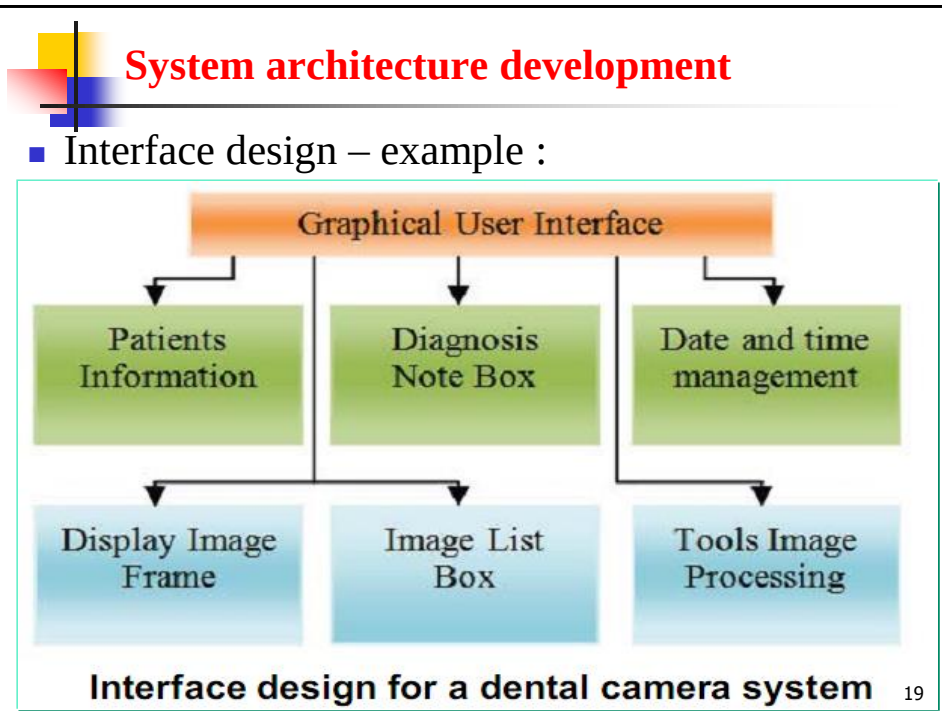


Database

14



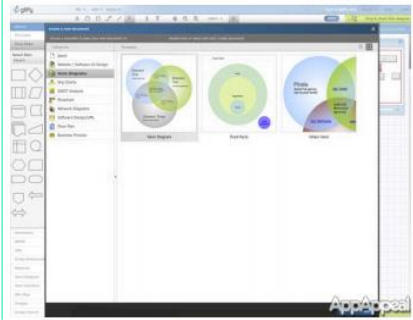
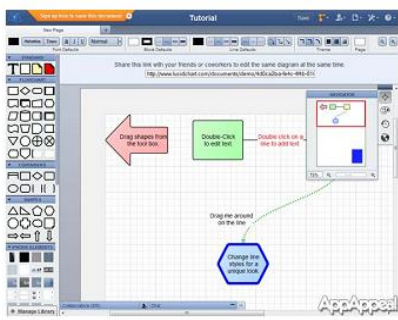




19

## System architecture development

- Diagram software tools
  - Microsoft Visio
  - Gliffy
  - Lucid chart

20



## Integration and Testing

- Integration process
  - Configuration
    - Define configurations, parameters, versions
  - Hardware integration
    - integrate microprocessor, peripherals, sensors, actuators, user interfaces
  - Software integration
    - integrate OS, drivers, functions,
  - System integration

21



## Integration and Test

- Testing process
  - Hardware testing
    - CPU
    - Peripheral interface
    - Sensor
    - Actuator
    - User interface
  - Software testing
    - Operating system, drivers
    - Applications/functions
    - User interface
  - Hard/soft co-testing
    - Test cases

22



## Embedded System Analysis

- 1. Technology selection
- 2. System optimization
- 3. Cost analysis

23



## Technology Selection

- Technology
  - A manner of accomplishing a task, especially using technical processes, methods, or knowledge
- Three key technologies for embedded systems
  - Processor technology: general-purpose, application-specific, single-purpose
  - IC technology: Full-custom, semi-custom, PLD
  - Design technology: Compilation/synthesis, libraries/IP, test/verification

24

## Technology Selection

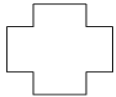
- Processor selection
  - number of IO pins required
  - interface required
  - memory requirements
  - number of interrupts required
  - real-time considerations
  - development environment
  - processing speed required



General-purpose processor



Application-specific processor



Single-purpose processor <sup>25</sup>

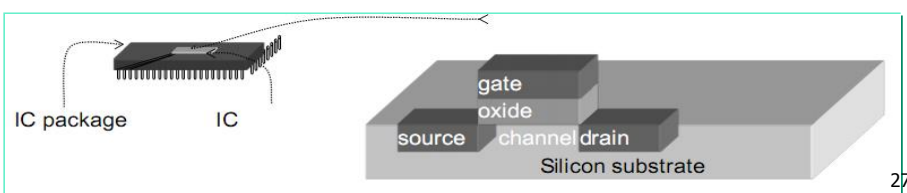
## Processor technology

- “Processor” not equal to general-purpose processor

| General-purpose (“software”)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Application-specific                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Single-purpose (“hardware”) <sup>26</sup>                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="border: 1px solid black; padding: 5px;"> <b>Controller</b><br/>           Control logic and State register<br/>           IR    PC         </div> <div style="border: 1px solid black; padding: 5px; margin-top: 5px;"> <b>Datapath</b><br/>           Register file<br/>           General ALU         </div> <div style="border: 1px solid black; padding: 5px; margin-top: 5px;">           Program memory<br/>           Assembly code for:<br/> <pre>total = 0 for i = 1 to ...</pre> </div> | <div style="border: 1px solid black; padding: 5px;"> <b>Controller</b><br/>           Control logic and State register<br/>           IR    PC         </div> <div style="border: 1px solid black; padding: 5px; margin-top: 5px;"> <b>Datapath</b><br/>           Registers<br/>           Custom ALU         </div> <div style="border: 1px solid black; padding: 5px; margin-top: 5px;">           Program memory<br/>           Assembly code for:<br/> <pre>total = 0 for i = 1 to ...</pre> </div> | <div style="border: 1px solid black; padding: 5px;"> <b>Controller</b><br/>           Control logic<br/>           State register         </div> <div style="border: 1px solid black; padding: 5px; margin-top: 5px;"> <b>Datapath</b><br/>           index<br/>           total<br/>           +         </div> <div style="border: 1px solid black; padding: 5px; margin-top: 5px;">           Data memory         </div> |

## IC technology

- The manner in which a digital (gate-level) implementation is mapped onto an IC
  - IC: Integrated circuit, or “chip”
  - IC technologies differ in their customization to a design
  - IC's consist of numerous layers (perhaps 10 or more)
    - IC technologies differ with respect to who builds each layer and when



The diagram illustrates the physical components of an integrated circuit. On the left, an 'IC package' is shown with pins, connected to an 'IC' chip. The chip is mounted on a 'Silicon substrate'. A cross-sectional view of the substrate shows several layers: 'source', 'gate oxide', 'channel', and 'drain'.

27

## System Optimization

- Hardware / software partitioning
  - which functions should be performed in hardware, and which in software?
  - the more functions in software, the lower will be the product cost

28



## System Optimization

Design challenge – optimizing design metrics

### ■ Common metrics

- 1. Unit cost: the monetary cost of manufacturing each copy of the system, excluding NRE cost
- 2. NRE cost (Non-Recurring Engineering cost): The one-time monetary cost of designing the system
- 3. Size: the physical space required by the system
- 4. Performance: the execution time or throughput of the system
- 5. Power: the amount of power consumed by the system
- 6. Flexibility: the ability to change the functionality of the system without incurring heavy NRE cost

29



### ■ Common metrics (continued)

- 7. Time-to-prototype: the time needed to build a working version of the system
- 8. Time-to-market: the time required to develop a system to the point that it can be released and sold to customers
- 9. Maintainability: the ability to modify the system after its initial release
- 10. Correctness: check the functionality throughout the process of designing the system; insert test circuitry to check that manufacturing was correct
- 11. Safety: the probability that the system will not cause harm

30



## The performance design metric

- Widely-used measure of system, widely-abused
  - Clock frequency, instructions per second – not good measures
  - Digital camera example – a user cares about how fast it processes images, not clock speed or instructions per second
- Latency (response time)
  - Time between task start and end
  - e.g., Camera's A and B process images in 0.25 seconds
- Throughput
  - Tasks per second, e.g. Camera A processes 4 images per second
  - Throughput can be more than latency seems to imply due to concurrency, e.g. Camera B may process 8 images per second (by capturing a new image while previous image is being stored).
- Speedup of B over S = B's performance / A's performance
  - Throughput speedup =  $8/4 = 2$

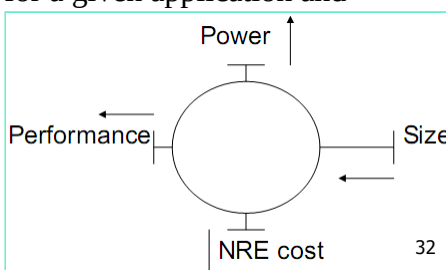
31



## System Optimization

Design metric competition -- improving one may worsen others

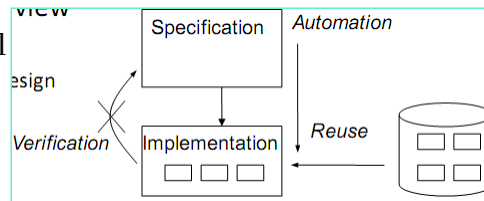
- Expertise with both software and hardware is needed to optimize design metrics
  - Not just a hardware or software expert, as is common
  - A designer must be comfortable with various technologies in order to choose the best for a given application and constraints



32

## Improving productivity

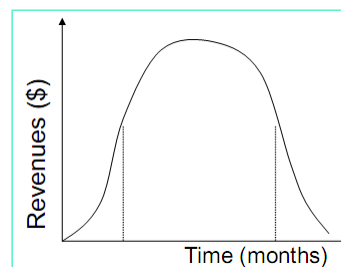
- Design technologies developed to improve productivity
- We focus on technologies advancing hardware/software unified view
  - Automation
    - Program replaces manual
    - Synthesis
  - Reuse
    - Predesigned components
    - Cores
    - General-purpose and single-purpose processors on single IC
  - Verification
    - Ensuring correctness/completeness of each design step
    - Hardware/software co-simulation



33

## Cost Analysis

- Time required to develop a product to the point it can be sold to customers
- Market window
  - Period during which the product would have highest sales
- Average time-to-market constraint is about 8 months
- Delays can be costly



34

